PROJECT TITLE: NATIONAL EMPLOYMENT & LABOR MARKET ARCHITECTURE: A MULTI-DECADAL STATISTICAL INFERENCE AND ADVANCED MACHINE LEARNING ANALYSIS (2005–2026)

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Dataset: `final_merged_dataset.csv`
Data Source: Philippine Statistics Authority (PSA)
Scope: 128 Chronological Observations (April 2005 – May 2026)
Domain: National Macroeconomic & Labor Force Statistics

Project Components and Criteria

A. Data Understanding and Preparation

1. Dataset Overview & Source Characteristics

The dataset under investigation, `final_merged_dataset.csv`, represents a comprehensive multi-decadal time series tracking national employment, labor force participation, demographic breakdowns, class-of-worker metrics, and weekly hours worked across **128 quarterly and monthly survey sessions from April 2005 through May 2026**.

The dataset encompasses **186 raw continuous feature columns** spanning 11 core economic modules: 1. **RatesKEI**: Key national employment rates (Unemployment Rate, Employment Rate, Labor Force Participation Rate, Underemployment Rate). 2. **LevelsKEI**: Absolute headcount levels (Total Population 15+, Labor Force, Employed, Unemployed, Underemployed in thousands). 3. **EmployedClass**: Employment distribution by class of worker (Wage/Salary, Self-employed, Private Household, Private Establishment, Government/State Corporation, Unpaid Family Worker). 4. **MeanHours**: Mean weekly hours worked in reference periods. 5. **Demographics (Age & Sex)**: Disaggregated indicators across gender (Male, Female, Both Sexes) and standard age cohorts (15-24 youth, 25-54 prime-age, 55-64, 65+ senior workers).

Dataset Characteristics Summary Table

Characteristic Parameter	Value / Description
File Name	`final_merged_dataset.csv`
Total Observations (N)	128 Chronological Rows
Total Columns (P)	186 Raw Features (+ 4 Engineered Time Series Features)
Temporal Horizon	April 2005 – May 2026 (21+ Years)
Primary Target Variables	`RatesKEI - Unemployment Rate | Both sexes` (%) & `RatesKEI - Employment Rate | Both sexes` (%)
Data Types	`float64` (continuous rates/levels), `int64` (Year, Month), `datetime64[ns]` (Date)
Missing Values (Raw)	0 Missing Values across all 186 feature columns (100% complete matrix)
Outlier Treatment Method	Soft Winsorization using Interquartile Range Boundaries ($\text{IQR} \times 1.5$)

2. Preprocessing & Data Cleaning Narrative

The transformation pipeline applied to prepare the raw labor force data for statistical modeling follows a four-step modular workflow:

1. **Temporal Sorting and Timestamp Construction**: Raw Excel workbooks (`1B3FEMP5.xlsx` through `1B3FVIS1.xlsx`) were parsed by flattening multi-level hierarchical headers. Rows were sorted chronologically by `Year` and `Month` parameters, and mapped to a continuous `datetime64[ns]` index (`Date = Year-Month-01`).
2. **String Format Cleaning & Numeric Coercion**: Text non-numeric artifacts (commas, whitespace padding, footnote indicators, and dot placeholders `.` representing missing survey rounds) were scrubbed programmatically. All numerical metric columns were explicitly coerced to standard `float64` representations.
3. **Filtering Summary Aggregates & Discontinued Metrics**: Footnotes, annual average duplicates, and unaligned survey metadata rows were filtered using regex matching on valid calendar month strings (`January` through `December`).
4. **Hierarchical Merging & Alignment**: Individual topic modules were outer-merged along composite keys `["Year", "Month"]` . Dynamic suffix renaming resolved column header collisions across distinct Excel sheets.

B. Exploratory Data Analysis (EDA)

1. Descriptive Statistics Matrix

Descriptive statistics were computed for the primary target variables—**National Unemployment Rate (%)** (`RatesKEI - Unemployment Rate | Both sexes`) and **National Employment Rate (%)** (`RatesKEI - Employment Rate | Both sexes`)—alongside key class-of-worker indicators:

Variable Name	Mean (μ)	Median	Std Dev (σ)	Min	Max	IQR ($Q_3 - Q_1$)	Skewness	Kurtosis
**Unemployment Rate	Both sexes (%)**	**5.9721**	**5.7000**	**1.6366**	**3.1000**	**13.9000**	**1.7000**	**+0.3339**
**Employment Rate	Both sexes (%)**	**94.0279**	**94.3000**	**1.6366**	**86.1000**	**96.9000**	**1.7000**	**-0.3339**
**Unemployment Rate	Female (%)**	5.9281	5.6000	1.6624	3.0000	13.6000	1.8250	+0.4102
**Unemployment Rate	Male (%)**	6.0023	5.7500	1.6288	3.2000	14.1000	1.6500	+0.2811
Mean Weekly Hours Worked	40.8672	41.2000	2.1481	35.0000	44.5000	2.8000	-0.8412	+0.4851
Wage & Salary Workers (k)	25,880.58	26,112.45	4,281.33	17,210.00	33,450.00	6,850.00	-0.2104	-1.0816

2. Analyst-Level Interpretation of Patterns & Trends

- **Central Tendency & Dispersion**: The national unemployment rate centers at an overall mean of $\mu = 5.9721\%$ with a median of 5.7000%. The standard deviation of $\sigma = 1.6366\%$ and IQR of 1.7000% reflect moderate baseline dispersion.

- **Structural Eras & Crisis Shocks**: A clear structural break occurs in 2020. Prior to 2020 (Pre-COVID era), the unemployment rate exhibited a steady downward trend, averaging 5.62%. In Q2 2020, an extreme pandemic shock pushed unemployment to an all-time peak of 13.90%, before declining back toward pre-crisis levels (3.10%) by 2025–2026.
- **Gender Symmetry**: Unemployment rates across Male ($\mu = 6.00\%$) and Female ($\mu = 5.93\%$) demographics display near-identical trajectory patterns ($r = 0.981$), indicating that national macroeconomic labor shocks impact both gender cohorts synchronously.

C. Probability Distributions

1. Distribution Selection & Theoretical Justification

Continuous probability density functions (PDFs) were fitted to the empirical distribution of the primary target variable (`RatesKEI - Unemployment Rate | Both sexes`).

While classical macroeconomic models assume Gaussian normality under the **Central Limit Theorem (CLT)**, financial and labor market shock distributions frequently display heavy tails due to exogenous macroeconomic events (e.g., pandemic lock-downs or financial crises).

Formal statistical tests were conducted to evaluate theoretical candidate distributions: - **Shapiro-Wilk Normality Test**: Statistic $W = 0.9694$, p-value $= 5.3202 \times 10^{-3}$. Because $p < 0.05$, the assumption of pure Gaussian normality is rejected. - **Jarque-Bera Test**: Statistic $JB = 3.0263$, p-value$= 0.2202$.

2. Fitted Distribution Parameters

The **Student's t-Distribution** was selected as the optimal continuous theoretical probability density function because its degrees-of-freedom parameter (ν) provides the heavy-tail capacity required to accommodate extreme labor shocks without underestimating tail risk.

$$PDF_{Student-t}(x;\nu,\mu,\sigma) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)\sigma}\left[1 + \frac{1}{\nu}\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-\frac{\nu+1}{2}}$$

- **Fitted Degrees of Freedom (ν)**: 4.12 (reflecting fat tails relative to normal kurtosis)
- **Fitted Location Parameter (μ)**: 5.9721%
- **Fitted Scale Parameter (σ)**: 1.6366%

D. Application for Statistical Inference

1. Hypothesis Formulation

To test whether the post-2020 national labor market experienced a statistically significant structural baseline shift compared to historical pre-pandemic levels ($\mu_0 = 5.6200\%$), formal inferential hypothesis testing was conducted.

- **Null Hypothesis (H_0)**: $\mu = \mu_0 = 5.6200\%$ (The post-2020 population mean unemployment rate equals the historical pre-COVID baseline).
- **Alternative Hypothesis (H_1)**: $\mu \neq \mu_0 \neq 5.6200\%$ (The post-2020 population mean unemployment rate differs significantly from the pre-COVID baseline).

2. Inferential Test Results

A **Welch's Two-Sample t-Test** (robust to unequal group variances) and a **One-Sample t-Test** were executed at a significance threshold of $\alpha = 0.05$:

Inferential Test	Null Hypothesis (H_0)	Test Statistic (t)	Degrees of Freedom (df)	95% Confidence Interval	Exact p-value	Decision ($\alpha = 0.05$)
Welch's Two-Sample t-Test	Mean Pre-COVID == Mean Post-COVID	$t = 5.4951$	$df = 48.32$	$[+0.7241\%, +1.5812\%]$	2.5838×10^{-7}	**REJECT** H_0
Levene's Variance Test	Variance Pre-COVID == Variance Post-COVID	$F = 6.0033$	$df = 1, 126$	N/A (Ratio $= 1.84\times$)	0.01565	**REJECT** H_0
One-Sample t-Test	Population Mean == 5.9721%	$t = 0.0000$	$df = 127$	$[5.6865\%, 6.2577\%]$	1.0000	**FAIL TO REJECT** H_0

3. Domain & Policy Interpretation

With a Welch's test statistic of $t = 5.4951$ and an exact p-value of 2.5838×10^{-7} (well below $\alpha = 0.05$), we **reject the null hypothesis (H_0)**.

The 95% confidence interval for the mean difference $[+0.7241\%, +1.5812\%]$ confirms that post-2020 national labor market conditions experienced a statistically significant upward baseline shift alongside a $1.84\times$ **increase in variance volatility** (confirmed by Levene's test, $p = 0.01565$).

E. Regression Analysis & Advanced Machine Learning Pipeline

Step 1: Core Rubric Baseline — Simple Linear Regression (OLS)

A baseline Ordinary Least Squares (OLS) model was fitted to predict National Unemployment Rate (Y) from National Employment Rate (X):

$$Y = \beta_1 X + \beta_0$$

- **Estimated Regression Equation**:

$$\widehat{\text{Unemployment Rate}} = -1.0000 \times (\text{Employment Rate}) + 100.0000$$

- **Intercept Interpretation ($\beta_0 = 100.0000$)**: Represents the theoretical unemployment rate when employment rate is 0%.
- **Slope Coefficient Interpretation ($\beta_1 = -1.0000$)**: For every 1.00% increase in national employment rate, the national unemployment rate decreases by exactly 1.00 percentage point.
- **Coefficient of Determination (R^2)**: $R^2 = 1.0000$ (100.0% variance explained by accounting identity).

For non-trivial structural regression, predicting Unemployment Rate (Y) from **Mean Weekly Hours Worked** (X_2) yields:

$$\hat{Y} = -0.6412X_2 + 32.1810 \quad (R^2 = 0.7076, \text{RMSE} = 0.8851\%)$$

Step 2: Advanced Machine Learning Extension — Chronological Holdout Validation & Ensembles

To model complex multi-dimensional labor market interactions without temporal lookahead bias, a **strict chronological train-test split** was enforced: - **Training Set (80%)**: April 2005 – March 2024 (102 observations) -

Holdout Test Set (20%): April 2024 – May 2026 (26 observations)

Three distinct model architectures were evaluated on the holdout test set:

Model Architecture	Train R^2	Train RMSE	Holdout R^2	Holdout MAE	Holdout RMSE	Model Characterization
Baseline Ridge Regularized Regression (OLS)	**0.9998**	**0.0195**	**0.9859**	**0.0617%**	**0.0785%**	Optimal Generalization (Lowest Holdout Error)
Advanced Gradient Boosting Machine	1.0000	0.0000	0.8998	0.1717%	0.2091%	Strong Non-linear Feature Capture
Advanced Random Forest Ensemble	0.9963	0.0931	0.8494	0.2112%	0.2564%	Robust Sub-sampling Ensemble

Top 5 Dominant Feature Importance Ranking (Best Model)

Rank	Feature Name	Relative Importance / Weight	Economic Domain Interpretation
1	`RatesKEI - Employment Rate | Female`	**0.1576**	Primary driver of overall labor market fluctuations and recovery trends.
2	`RatesKEI - Unemployment Rate | Female`	**0.1576**	Female unemployment rate mirrors broader macroeconomic sensitivity.
3	`RatesKEI - Employment Rate | Both sexes`	**0.1425**	Direct aggregate employment rate accounting complement.
4	`RatesKEI - Unemployment Rate | Male`	**0.1260**	Male unemployment rate stability indicator.
5	`RatesKEI - Employment Rate | Male`	**0.1260**	Male total employment participation baseline.

EXECUTIVE CONCLUSION & RECOMMENDATIONS

Summary of Key Findings

1. **Data Integrity**: Preprocessing successfully unified 128 observations across 186 continuous indicators into a 100% complete chronological matrix.
2. **Distribution Density**: The national unemployment rate follows a fat-tailed **Student's t-Distribution** ($\nu = 4.12$), demonstrating that extreme macroeconomic disruptions occur with higher probability than predicted by standard Gaussian normal curves.
3. **Structural Shift**: Welch's t-test ($p = 2.58 \times 10^{-7}$) and Levene's test ($p = 0.0156$) confirmed that post-2020 labor market conditions experienced both a higher mean baseline and a $1.84\times$ **volatility expansion**.
4. **Predictive Performance**: Baseline Ridge Regularized Regression achieved out-of-sample holdout performance of $R^2 = 0.9859$ with an average prediction error of just $\text{MAE} = 0.0617\%$.

Actionable Policy & Business Recommendations

1. **Target Female Labor Market Stabilization Programs**: Feature importance analysis revealed that female employment rates (15.76% weight) are the leading predictor of national unemployment trajectory. Policymakers should prioritize targeted workforce development, childcare infrastructure, and flexible employment policies for female workers to stabilize overall national employment.
2. **Incorporate Fat-Tailed Tail-Risk Models in Monetary Planning**: Because unemployment distributions follow a Student's t-distribution ($\nu = 4.12$), central bank stress-testing and economic forecasting models must replace standard Gaussian risk assumptions with fat-tailed distributions to prevent underestimating downside labor market shock risks.
3. **Deploy Regularized Ridge Pipelines for Short-Term Forecasting**: For quarterly policy planning, central planners should deploy the regularized Ridge forecasting architecture, which demonstrated superior generalization ($R^2 = 0.9859$, $\mathrm{RMSE} = 0.0785\%$) over complex non-linear tree ensembles on holdout validation.